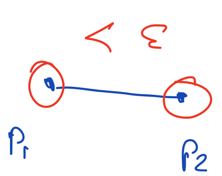
Last Class (Oct 13)

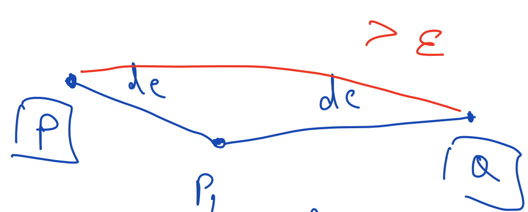
- Gaussian Mixture Models

Today's agenda

1) Case study : customer grouping
2) DBSCAN : key ideas
3) DBSCAN Algorithm
4) Adjusting params
5) Advantages & Disadvantages of DBSCAN
6) DBSCAN implementation
7) Intro to Anomaly / Novelty / Outlier Detection
8) Business Case Intro
9) RANSAC & Elliptical Envelope

Problem Solving Session





density connected points

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad . \quad x_6$

$x_1 \quad x_2 \quad d_{12} \quad v_1$

$v_1 < v_2 < v_3 < \dots < v_6$

$x_1 \quad x_3 \quad d_{13} \quad v_2$

$x_1 \quad x_4 \quad d_{14} \quad v_3$

$\boxed{v_4}$

Sort them in ascending manner

$\vdots$

$d_{16} \quad v_6$

$x_1 \longrightarrow v_{1,4}$ ✓ 4th NN distance

$x_2 \longrightarrow v_{2,4}$

$x_3 \longrightarrow v_{3,4}$

Sort $v_{1,4} \quad v_{2,4} \quad \dots \quad v_{N,4}$

$\vdots$

$x_N \longrightarrow v_{N,4}$

Loss fn = K-Means

$$\sum_{\substack{i=1 \\ x_i \in C_j}}^{N} |x_i - c_j|^2$$

GMM

$$\sum_{i=1}^{N} w_1 |x_i - c_1|^2 + w_2 |x_i - c_2|^2 + \dots$$

$$\boxed{w_1 + w_2 + \cdots w_{12} = 1}$$

$\hookrightarrow$ prob.

$w_1, w_2, \ldots W_{10}$

$\hookrightarrow 1 \longrightarrow 0$